Software Requirements Specification (SRS)

Project: Electronic Store Inventory Management System | **Version:** 1.0 | **Authors:** Karnati Uday Keerthan Reddy, K B Balavideshna, Kolluri Sruthi | **Status:** Approved

Revision History

Version	Author	Change Summary	Approval
1.0	Karnati Uday Keerthan Reddy, K B Balavideshna, Kolluri Sruthi	Complete SRS consolidating functional, non-functional, security requirements, use-case diagrams, use-case flows, and RTM.	Approved

Approvals

Role	Name	Signature / Status
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Table of Contents

- 1. Introduction (1.1 Purpose, 1.2 Scope, 1.3 Audience, 1.4 Definitions)
- 2. Overall Description (2.1 Product Perspective, 2.2 Major Product Functions, 2.3 User Roles & Characteristics, 2.4 Operating Environment, 2.5 Constraints)
- 3. External Interface Requirements (3.1 User Interfaces, 3.2 Hardware Interfaces, 3.3 Software Interfaces, 3.4 Communications)
- 4. System Features — Detailed Functional Requirements (FR-001 to FR-015)
- 5. Non-Functional Requirements — Detailed (NFR-001 to NFR-012) & 5.1 Security (Objectives & SEC-001 to SEC-016)
- 6. Quality Attributes & Acceptance Tests
- 7. System Models and Diagrams (7.1 UML Use-Case Diagrams, 7.2 Detailed Use-Case Flows)
- 8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, and security requirements for the Electronic Store Inventory Management System. It establishes verifiable acceptance criteria, external interface definitions, system use-case models, and end-to-end requirement traceability for development, testing, and academic evaluation.

1.2 Scope

The system is a web-based inventory, point-of-sale (POS) billing, procurement, and reporting application designed for a retail electronics store. The scope covers role-based user authentication (Admin, Manager, Staff), product catalog CRUD with unique SKU enforcement, multi-item sales checkout with 5% tax and itemized invoice generation, automatic stock deduction and customer returns/refunds, configurable low-stock alerts, supplier management, purchase order creation and full/partial delivery receipt, selectable date-range reporting (sales, stock movements, and revenue) with CSV/PDF export, immutable audit logging, and automated SQLite database backups. External payment gateway card processing and multi-branch ERP synchronization are outside the scope of this system.

1.3 Audience

Software developers, quality assurance (QA) and testing engineers, security auditors, store operations stakeholders, and course assessment evaluators.

1.4 Definitions

Term / Acronym	Definition
SKU	Stock Keeping Unit — unique alphanumeric identifier assigned to each hardware product in inventory.
POS	Point of Sale — retail checkout workflow where staff record customer purchases and generate invoices.
RBAC	Role-Based Access Control — server-side permission model restricting routes across Admin, Manager, and Staff roles.
PO	Purchase Order — formal replenishment order issued by a Manager or Admin to an approved Supplier.
ACID	Atomicity, Consistency, Isolation, Durability — database transaction properties preventing partial updates.
RPO	Recovery Point Objective — maximum acceptable data loss window (24 hours via automated daily backups).
TLS / HTTPS	Transport Layer Security / Hypertext Transfer Protocol Secure — encrypted network communication protocol.
RTM	Requirements Traceability Matrix — mapping linking SRS requirements to design modules and verification test cases.

2. Overall Description

2.1 Product Perspective

The Electronic Store Inventory Management System is a self-contained, three-tier layered web application comprising a Presentation Layer (Flask blueprints, Jinja2 templates, and RBAC route guards), a Business Logic Layer (domain services for inventory management, sales/billing, returns, purchase orders, password security, audit logging, and report exports), and a Data Access Layer (SQLAlchemy ORM models and relational SQLite database storage), supported by an operational backup utility.

2.2 Major Product Functions (Detailed)

- **Authenticate & Authorize Users:** Verify user credentials against salted bcrypt hashes, manage active sessions, support token-based password recovery, and enforce role-based access control (Admin, Manager, Staff).
- **Product Catalog & SKU Management:** Add, edit, and delete electronics products (name, SKU, category, unit price, stock quantity, reorder threshold, and linked supplier) with duplicate-SKU rejection and sales-history deletion safeguards.
- **Search & Filter Inventory:** Lookup products by name or SKU and filter by hardware category and stock status (In Stock, Low Stock, Out of Stock).
- **Process Sales & Generate Invoices:** Validate stock across single or multi-item carts, compute subtotals, 5% tax, and grand totals using exact decimal arithmetic, atomically deduct stock, and render printable invoices (INV-<id>).
- **Process Customer Returns & Refunds:** Record product returns against completed sales or standalone items, restore returned quantities to shelf stock, and log calculated refund amounts.
- **Low-Stock Alert Monitoring:** Automatically detect products whose stock falls below their configured reorder threshold and display actionable dashboard alerts.
- **Supplier & Purchase Order Lifecycle:** Maintain supplier records, generate purchase orders manually or directly from low-stock alerts, and process full or partial deliveries that automatically increment product stock.
- **Operational & Financial Reporting:** Generate Sales, Stock Movement, and Revenue reports over selectable date ranges and export report data to CSV and PDF files.
- **Audit Trail & Database Backups:** Log critical CRUD, stock, and price modifications with user ID and timestamp, and execute automated daily SQLite backups verified via PRAGMA integrity_check.

2.3 User Roles and Characteristics (Expanded)

User Role	Responsibilities & Permitted Operations	Access Restrictions
Staff	Search and filter products, process customer sales transactions, view/print sales invoices, and process product returns/refunds.	Cannot manage users, suppliers, or product deletions; cannot create purchase orders or access management reports (HTTP 403).
Manager	All Staff capabilities plus viewing low-stock alerts, creating purchase orders, marking purchase orders as received or partially delivered, configuring reorder thresholds, and generating/exporting sales, stock, and revenue reports.	Cannot manage user accounts, create/edit suppliers, or delete catalog products (HTTP 403).

User Role	Responsibilities & Permitted Operations	Access Restrictions
Admin	Full system administration: manage user accounts and roles (create, edit, deactivate, reactivate), manage supplier records, perform full product CRUD including product deletion, and access all purchase order and reporting workflows.	Self-deactivation of the currently logged-in Admin account is blocked to prevent administrative lockout.

2.4 Operating Environment

Operating System: macOS and Windows (both were used) for development, automated testing, and local store deployment. **Runtime & Frameworks:** Python 3.14.6, Flask 3.1.3, Flask-Login 0.6.3, SQLAlchemy 2.1.6, bcrypt 5.0.0, ReportLab 5.1.0. **Database Engine:** SQLite 3 relational database (file-backed in production/development and isolated in-memory/temporary file instances in automated testing). **Client Environment:** Modern standards-compliant web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge) without platform-specific plugins.

2.5 Constraints

All database access must occur through parameterized SQLAlchemy ORM queries; passwords must never be stored or logged in plaintext; financial calculations must use exact Decimal arithmetic to prevent floating-point rounding errors; and multi-item sales or restock operations must execute as atomic database transactions.

3. External Interface Requirements

3.1 User Interfaces

The system provides a responsive browser-based user interface rendered via modular Jinja2 templates and CSS stylesheets. Core views include the POS & Inventory Dashboard (/), Product Creation & Editing forms (/products/create, /products/<id>/edit), Invoice Receipt view (/sales/<id>/invoice) with browser print trigger, Purchase Order Management views (/purchase-orders, /purchase-orders/create, /purchase-orders/<id>), Supplier Management views (/suppliers, /suppliers/create, /suppliers/<id>/edit), Date-Range Report views (/reports/sales, /reports/stock, /reports/revenue), and Authentication/User Administration views (/login, /forgot-password, /reset-password/<token>, /users).

3.2 Hardware Interfaces

The application executes on standard retail computer workstations running macOS and Windows (both were used) and interfaces with standard display monitors, keyboard/barcode-wedge input devices, and OS-connected receipt/document printers via the browser's native window.print() interface.

3.3 Software Interfaces

Database Interface: SQLAlchemy 2.1.6 ORM over SQLite3 connection pool. **Email Recovery Interface:** Outbound password-reset notification service (app/business/password_reset_email.py:send_reset_email) dispatching time-limited signed reset URLs. **Report Export Interface:** CSV writer and ReportLab PDF document generator (app/business/report_exporter.py). **Backup Interface:** SQLite native backup API and PRAGMA integrity_check verifier (scripts/backup_database.py).

3.4 Communications

Client-server communication uses HTTP/1.1 and HTTPS (TLS 1.2+ in production deployment). Session state is maintained via signed, HttpOnly session cookies, and password reset workflows use cryptographically signed itsdangerous URLSafeTimedSerializer tokens with a 30-minute expiration window.

4. System Features (Detailed Functional Requirements)

This section specifies all 15 Functional Requirements (FR-001 through FR-015), grouped by functional subsystem. FR-001 through FR-012 preserve the project's baseline SRS specification verbatim, and FR-013 through FR-015 document the implemented partial purchase order delivery, report CSV/PDF export, and user account lifecycle features to satisfy the template minimum of 15 functional requirements.

4.1 Product Catalog, Inventory Synchronization & Search (FR-001, FR-002, FR-003, FR-008)

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria & Test Ref	Rationale / Dependencies
FR-001	System shall allow authorized users to add, update, and delete product records (name, SKU, category, price, quantity, supplier).	Functional	High	Product CRUD operations succeed with valid input; duplicate SKU rejected. Refs: IT_03, IT_04, test_products.py.	Accurate product data is the foundation for all inventory operations.
FR-002	System shall automatically update stock quantity on sale, return, or restock.	Functional	High	Stock count reflects correct value within 2 sec of transaction. Refs: UT_01, IT_07, IT_08, IT_11, test_sales.py.	Prevents overselling and manual reconciliation errors.
FR-003	System shall generate an alert when stock falls below a predefined reorder threshold.	Functional	High	Alert triggers immediately when threshold condition met. Refs: ST_01, test_products.py.	Enables timely restocking and avoids stockouts.
FR-008	System shall allow users to search and filter products by name, SKU, category, or stock status.	Functional	Medium	Search returns correct matching results within 1 sec. Refs: test_products.py, test_main_routes_coverage.py.	Improves usability during high-volume checkout or lookup.

4.2 Point-of-Sale Billing, Invoices & Customer Returns (FR-004, FR-011)

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria & Test Ref	Rationale / Dependencies
FR-004	System shall allow staff to create a sale transaction and generate an invoice.	Functional	High	Invoice generated with itemized list, total, date, transaction ID (INV-<id>). Refs: UT_01, UT_02, IT_07, test_sales.py.	Core business function; required for customer transactions and record-keeping.
FR-011	System shall allow staff to process returns/refunds, updating stock and transaction records.	Functional	Medium	Returned item quantity added back to stock; refund recorded. Refs: IT_11, test_sales.py.	Supports accurate inventory and financial reconciliation.

4.3 Authentication, Role-Based Access, Password Recovery & User Administration (FR-005, FR-010, FR-012, FR-015)

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria & Test Ref	Rationale / Dependencies
FR-005	System shall authenticate users and restrict actions based on role (Admin, Manager, Staff).	Functional	High	Unauthorized role access returns 'Access Denied' (HTTP 403 / redirect). Refs: IT_01, IT_02, IT_09, ST_03, test_auth.py.	Protects sensitive data and prevents unauthorized actions.
FR-010	System shall log key actions (stock edits, deletions, price changes) with timestamp and user ID.	Functional	High	Every CRUD action creates a corresponding audit log entry. Refs: IT_06, test_products.py.	Ensures accountability and supports dispute resolution.
FR-012	System shall allow password reset for users who are locked out of their accounts.	Functional	Low	User receives reset link/OTP and can set a new password successfully. Refs: test_auth.py, test_password_reset_email.py.	Reduces support overhead and downtime due to lost credentials.
FR-015	System shall allow admins to create, edit, deactivate, and reactivate user accounts while preventing self-deactivation.	Functional	Medium	Admin can create/edit/deactivate/reactivate accounts; deactivated users cannot log in; self-deactivation blocked. Refs: test_auth.py, test_auth_routes_coverage.py.	Enables complete staff lifecycle governance without risking admin lockout.

4.4 Supplier Management & Purchase Order Replenishment (FR-006, FR-007, FR-013)

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria & Test Ref	Rationale / Dependencies
FR-006	System shall allow admins to add/edit supplier details and link products to suppliers.	Functional	Medium	Supplier record saved; linked products display correct supplier. Refs: IT_09, test_suppliers.py.	Enables efficient reordering and supply chain tracking.
FR-007	System shall allow managers to create purchase orders and mark them as received.	Functional	High	PO created with 'Pending' status; stock updates on 'Received'. Refs: IT_08, test_purchase_orders.py.	Supports systematic restocking workflow.
FR-013	System shall allow managers and admins to record partial deliveries against pending purchase orders and generate purchase orders directly from low-stock alerts.	Functional	Medium	Partial delivery increments stock by delivered quantity, updates pending quantity, and sets status to 'Partially Received' or 'Received'. Refs: test_purchase_orders.py.	Handles split vendor shipments accurately and accelerates low-stock replenishment.

4.5 Analytical Reporting & Data Exports (FR-009, FR-014)

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria & Test Ref	Rationale / Dependencies
FR-009	System shall generate sales, stock, and revenue reports over a selectable date range.	Functional	Medium	Report totals match underlying transaction records for selected range. Refs: ST_02, ST_04, test_sales_report.py, test_stock_report.py, test_revenue_report.py.	Supports business decision-making and performance tracking.
FR-014	System shall allow managers and admins to export generated sales and stock movement reports in CSV and PDF formats.	Functional	Medium	Export endpoints return valid CSV (text/csv) and PDF (application/pdf) files matching the filtered date range. Refs: test_report_exports.py.	Enables offline archival, accounting review, and regulatory reporting.

5. Non-Functional Requirements (Detailed)

All 12 Non-Functional Requirements (NFR-001 through NFR-012) are measurable and mapped to verification criteria.

Req ID	Requirement (shall...)	Category	Priority	Acceptance Criteria / Measurement	Rationale
NFR-001	System shall process a sale transaction and update inventory within 2 seconds.	Performance	High	Transaction completion time ≤ 2 sec under normal load. Ref: UT_01, IT_07.	Ensures smooth checkout experience, avoids customer wait times.
NFR-002	System shall maintain 99% uptime during store operating hours.	Reliability / Availability	High	Uptime monitoring logs show ≥ 99% availability.	Store operations depend on continuous system access.
NFR-003	Passwords shall be stored using secure hashing (e.g., bcrypt); RBAC enforced on all endpoints.	Security / Access Control	High	No plaintext passwords in DB; restricted endpoints return 401/403 for unauthorized calls. Ref: IT_09, ST_03.	Protects against data breaches and unauthorized access.
NFR-004	UI shall allow a new staff member to complete a basic sale within 15 minutes of training.	Usability	Medium	New user completes sale transaction unaided within 15 min.	Reduces onboarding cost and training overhead.
NFR-005	System shall ensure atomic transactions so stock and billing records never desynchronize.	Data Integrity / Atomicity	High	No partial updates occur even if a transaction fails mid-way. Ref: IT_10.	Maintains data consistency and financial accuracy.
NFR-006	System shall support up to 10,000 product records and 50 concurrent users without performance degradation.	Scalability / Capacity	Medium	Load test with 50 concurrent users shows no crash or major slowdown.	Ensures system scales with business growth.
NFR-007	System shall be accessible via modern web browsers (Chrome, Firefox, Edge) without platform-specific installation.	Portability / Compatibility	Medium	UI renders and functions correctly across all three browsers.	Ensures accessibility across different store devices.
NFR-008	Codebase shall follow modular architecture (MVC / Layered) to allow independent module updates.	Maintainability	Medium	Inventory, billing, and reporting modules can be updated independently without breaking others.	Improves long-term maintainability and reduces regression risk.
NFR-009	System shall perform automated daily database backups with a 24-hour recovery point objective.	Backup / Recovery	High	Backup file generated daily and successfully restorable within 24 hrs. Ref: IT_12.	Protects against data loss and supports business continuity.
NFR-010	System shall retain audit logs for a minimum of 6 months.	Audit / Data Retention	Medium	Logs older than current date but within 6 months remain queryable. Ref: IT_06.	Supports compliance and long-term traceability.
NFR-011	System shall encrypt sensitive data (e.g., user credentials) in transit using HTTPS/TLS.	Transport Security	High	All API/browser communication occurs over HTTPS; no plaintext transmission detected.	Prevents interception of sensitive data over the network.
NFR-012	System shall provide clear, consistent error messages for all failed operations.	Usability / Error Handling	Low	Every failure case (invalid input, network error, etc.) shows a specific, user-understandable message. Ref: UT_02, IT_02, IT_04, IT_05.	Improves usability and reduces user confusion/support tickets.

5.1 Security

5.1.1 Security Objectives

- **SO-1 (Credential & Session Confidentiality):** Ensure that all user authentication credentials are salted and hashed using bcrypt (cost factor ≥ 10), sessions use HttpOnly cookies with inactivity timeouts, and network traffic is protected via TLS 1.2+.
- **SO-2 (Server-Side Least-Privilege Authorization):** Enforce deterministic Role-Based Access Control (Admin, Manager, Staff) on every server endpoint so that UI hiding is never relied upon as a security control.

- **SO-3 (Transactional & Audit Non-Repudiation):** Guarantee atomic database commits/rollbacks for sales and stock updates and maintain an immutable 6-month audit log of all critical state changes with actor user ID and timestamp.
- **SO-4 (Input Defense & Operational Continuity):** Validate all server inputs, execute parameterized ORM queries, encode rendered HTML outputs against XSS, avoid storing payment card data, and maintain verified daily database backups.

5.1.2 Security Requirements (SEC-001 to SEC-016)

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria	Rationale / Related Req
SEC-001	The system shall require every user to log in with a unique username and password before accessing any function other than the login and password-reset pages.	Security	High	Requests to any protected page or API without a valid session return 401 or redirect to login. No shared or anonymous accounts exist.	Related: FR-005. Establishes identity for all later checks.
SEC-002	The system shall enforce a password policy of at least 8 characters with upper-case, lower-case, a digit and a special character.	Security	Medium	A password that violates any rule is rejected with a message stating the unmet rule.	Reduces success of guessing attacks.
SEC-003	Passwords shall be stored only as salted hashes using bcrypt (cost factor 10 or higher). Plaintext or reversible storage is prohibited.	Security	High	Inspection of the users table shows only bcrypt hashes. Two users with the same password have different hashes.	Related: NFR-003. Limits damage if the database leaks.
SEC-004	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Security	High	The 6th attempt within the lockout window is refused even with the correct password. The lockout is recorded in the audit log.	Mitigates brute-force attacks.
SEC-005	Sessions shall expire after 30 minutes of inactivity, be invalidated on logout, and use tokens that are random, HttpOnly and sent only over HTTPS.	Security	High	A session idle for 30 minutes is rejected. A token reused after logout is rejected. Cookie flags HttpOnly and Secure are present.	Prevents session hijacking and unattended-terminal misuse at the checkout counter.
SEC-006	The system shall enforce role-based access control on the server for every request. Staff may search products and process sales and returns only. Managers add purchase orders and reports. Only Admins manage users, suppliers and product deletion.	Security	High	A Staff token calling a Manager or Admin endpoint returns 403 even when the UI hides the option. Role changes take effect on the next request.	Related: FR-005, NFR-003. UI-only hiding is not a control.
SEC-007	The application database account shall hold only the privileges it needs (read and write on application tables) and shall not be able to alter schema or access other databases.	Security	Medium	Attempting DROP or ALTER with the application credentials fails with a permission error.	Limits impact of an injection flaw or stolen application credentials.
SEC-008	All communication between browser and server shall use HTTPS with TLS 1.2 or higher. Plain HTTP requests shall be redirected to HTTPS.	Security	High	A capture of traffic shows no readable credentials. An HTTP request receives a redirect to HTTPS.	Related: NFR-011. Protects credentials and session tokens in transit.
SEC-009	All user input shall be validated on the server and all database access shall use parameterized queries. Output rendered in pages shall be encoded.	Security	High	Injection strings such as ' OR 1=1-- in login, search and product fields are rejected or stored as inert text. Script tags entered in a product name are displayed as text and do not execute.	Prevents SQL injection and cross-site scripting.

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria	Rationale / Related Req
SEC-010	The system shall log logins (success and failure), account lockouts, role changes, and every create, update or delete of products, stock, prices and purchase orders, recording user ID, timestamp, action and affected record.	Security	High	Each listed action produces one log entry with all fields. Logs cannot be edited or deleted through the application and are retained for at least 6 months.	Related: FR-010, NFR-010. Gives accountability and supports dispute resolution.
SEC-011	Password reset shall use a single-use token that expires in 15 minutes, delivered to the registered contact, and the response shall not reveal whether the account exists.	Security	Medium	A used or expired token is rejected. The reset form gives the same response for known and unknown usernames.	Related: FR-012. Prevents account takeover through the recovery path.
SEC-012	Error messages shown to users shall be generic and shall not expose stack traces, SQL text, file paths or software versions. Details shall be written to the server log only.	Security	Medium	Forcing a database error shows a generic message to the user while the detail appears in the server log.	Related: NFR-012. Denies attackers information about internals.
SEC-013	Stock changes and invoice records for a sale shall be committed in one atomic transaction, and invoices shall not be editable after creation. Corrections shall be made through a return or adjustment entry.	Security	High	A forced failure mid-sale leaves stock and billing unchanged. No function exists to edit a finalized invoice.	Related: NFR-005. Prevents silent alteration of financial records.
SEC-014	Daily database backups shall be encrypted and stored where only the Admin role and the backup service can read them. A restore test shall be performed at least once per semester.	Security	High	A backup file cannot be read without the key. Restoring the latest backup yields a working database.	Related: NFR-009. Protects availability and confidentiality of stored data.
SEC-015	The system shall not store card numbers, CVV or other payment card data. Invoices shall record only the payment method and amount.	Security	High	Database and log inspection finds no card data. Invoice records contain payment method type only.	Removes the most damaging data from the leak surface.
SEC-016	Third-party dependencies shall be listed in a software bill of materials and scanned for known vulnerabilities before each release. No known critical or high vulnerabilities shall remain unresolved at release.	Security	Medium	The dependency scan reports zero critical or high findings at release. The bill of materials is attached to the release.	Keeps known library flaws out of the product.

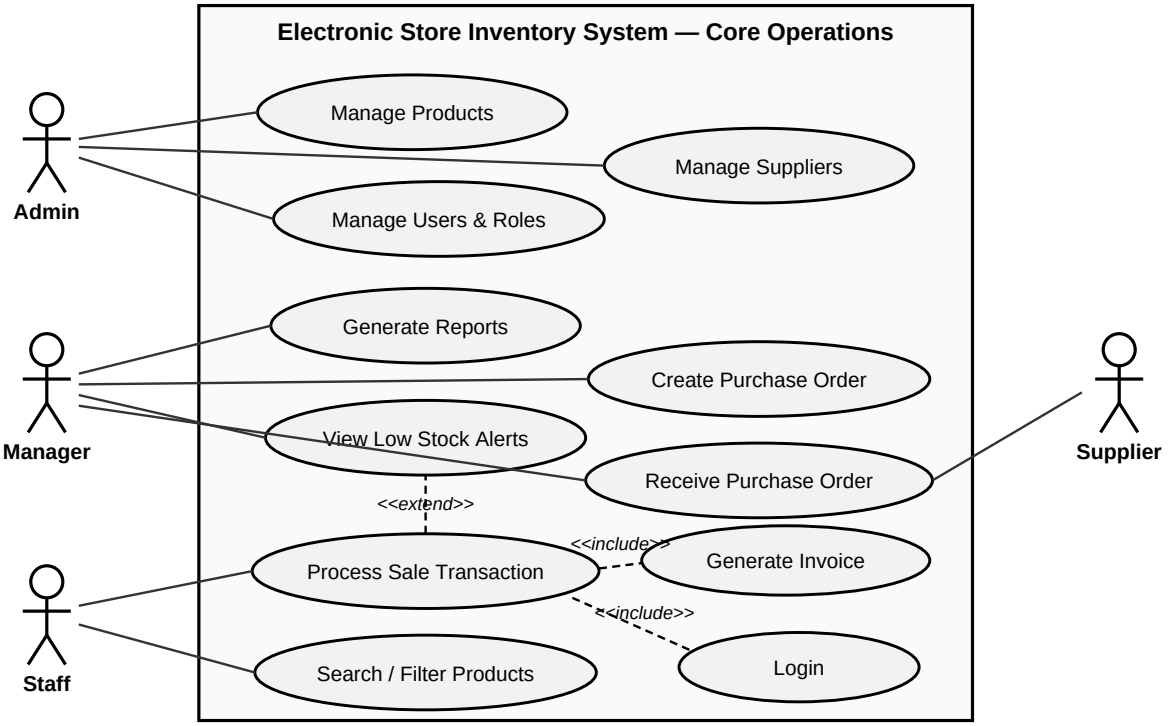
6. Quality Attributes & Acceptance Tests

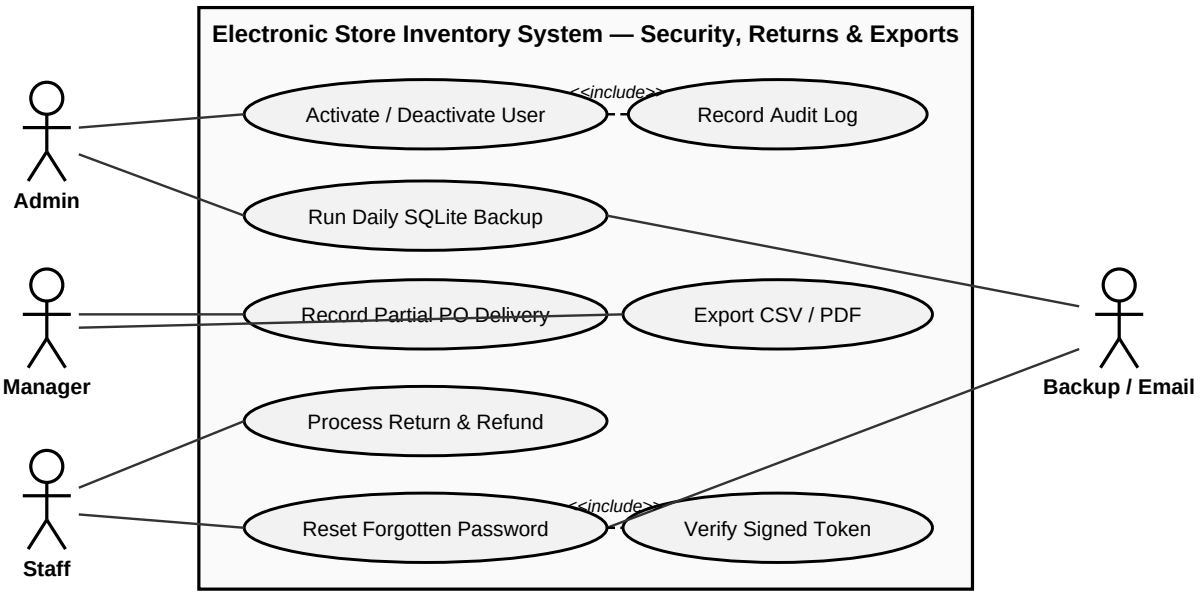
- **Exit Criteria for Acceptance:** All 18 formal system validation test cases (UT_01–UT_02, IT_01–IT_12, ST_01–ST_04) and all 373 automated Pytest tests across the 20 test modules must pass with 0 failures, achieving ≥ 90% line and branch coverage across the production codebase (actual: 97.70% line coverage, 91.50% branch coverage).
- **Acceptance Test Suites:** (1) Authentication, Session & RBAC Suite (test_auth.py, IT_01, IT_02, IT_09, ST_03); (2) Product Catalog, Search/Filter & Low-Stock Alert Suite (test_products.py, IT_03, IT_04, IT_05, ST_01); (3) Sales, Invoice, Returns & Atomic Rollback Suite (test_sales.py, UT_01, UT_02, IT_07, IT_10, IT_11); (4) Supplier & Purchase Order Replenishment Suite (test_suppliers.py, test_purchase_orders.py, IT_08); (5) Financial & Inventory Reporting & Export Suite (test_sales_report.py, test_stock_report.py, test_revenue_report.py, test_report_exports.py, ST_02, ST_04); and (6) Audit Logging & Database Backup Suite (IT_06, IT_12, test_backup_database.py).

7. System Models and Diagrams

7.1 UML Use-Case Diagrams

In accordance with the SRS template requirement (at least 2 UML use-case diagrams), Figure 7.1 models the core retail inventory, sales, purchase order, and reporting workflows, and Figure 7.2 models the security administration, password recovery, returns/refunds, partial delivery, report export, and backup workflows.





7.2 Detailed Use-Case Flows

Use Case / Goal	Actor	Precondition	Main Flow	Alternate Flow	Postcondition
UC-01: Process Sale Transaction	Staff	Staff is logged in, and the selected product exists in the inventory with sufficient stock.	1. Staff selects New Sale from the dashboard. 2. The system displays the product search screen. 3. Staff searches for and selects one or more products to add to the cart. 4. The system validates the stock availability for each selected item. 5. Staff confirms the required quantity for each item. 6. The system calculates the subtotal, tax, and final total. 7. Staff clicks Checkout. 8. The system deducts the sold quantity from the inventory stock. 9. The system generates an invoice containing the transaction ID, date, itemized product list, and total amount. 10. The system displays the invoice and provides an option to print it. 11. The sale transaction is completed.	Insufficient Stock (At Step 4): 4a. System displays an Insufficient Stock error message. 4b. System prevents the item from being added with the requested quantity. 4c. Staff corrects quantity before continuing to Step 5. | Low Stock Alert (At Step 11): 11a. System checks remaining stock level. 11b. If stock falls below configured reorder threshold, system triggers a low-stock alert. 11c. Manager is notified.	The inventory is updated, an invoice is generated, and an audit log entry is created.
UC-02: Manage Products	Admin	Admin is logged in.	1. Admin navigates to the Product Management screen. 2. System displays the list of existing products. 3. Admin chooses to add, edit, or delete a product. Add Product: Enter name, SKU, category, price, quantity, and supplier; system validates unique SKU and saves record. Edit Product: Select existing product, modify details, validate and update record. Delete Product: Select product, confirm deletion, remove record, record action in audit trail, and refresh product list.	Duplicate SKU (At Step 3 Add Product): 3a. Admin attempts to add a product using an existing SKU. 3b. System detects duplicate SKU. 3c. System displays an error message. 3d. Creation is cancelled until a unique SKU is provided.	The product catalog is updated, and an audit log entry is created.
UC-03: Manage Users and Roles	Admin	Admin is logged in.	1. Admin navigates to the User Management module. 2. Admin creates a new user account by entering username, password, and role (Admin, Manager, Staff). 3. System validates that the username is unique. 4. System assigns role-based permissions. 5. Admin can modify an existing user's role or deactivate/reactivate a user account. 6. System updates the user's access permissions immediately.	Username Already Exists (At Step 3): 3a. Admin enters a username that already exists. 3b. System rejects account creation and displays an error message.	The user account is created or updated with the correct role-based access permissions.
UC-04: Manage Suppliers	Admin	Admin is logged in.	1. Admin navigates to the Supplier Management module. 2. Admin enters supplier details (name, contact information, and address). 3. Admin links the supplier to one or more relevant products. 4. System saves the supplier record and updates product references.	Duplicate or Blank Supplier Name: System rejects submission and displays validation error message.	The supplier catalog is updated, and the supplier is linked to the relevant products.
UC-05: Create Purchase Order	Manager	Manager is logged in, and the required product and supplier records exist.	1. Manager views products with low stock or manually selects products. 2. Manager selects required product(s) and specifies ordered quantity. 3. Manager selects the appropriate supplier. 4. System generates a Purchase Order with status set to Pending. 5. Manager confirms and system saves the Purchase Order.	Invalid Quantity or Missing Supplier: System rejects the request and displays a validation error message.	A new Purchase Order is created with the status Pending.

Use Case / Goal	Actor	Precondition	Main Flow	Alternate Flow	Postcondition
UC-06: Receive Purchase Order	Manager	A Purchase Order with the status Pending or Partially Received exists.	1. Manager opens the Purchase Orders list and selects the relevant Purchase Order. 2. Manager marks the Purchase Order as Received. 3. System validates received quantity against ordered quantity. 4. System updates product stock according to received quantity. 5. System changes Purchase Order status to Received and logs the restock.	Partial Delivery (At Step 3): 3a. Supplier delivers only part of ordered quantity. 3b. Manager enters actual quantity received. 3c. System validates partial delivery, increases stock by received amount, and sets status to Partially Received while remaining quantity stays pending.	The inventory stock is increased according to the received quantity, and the Purchase Order status is updated.
UC-07: Generate Reports	Manager	Manager is logged in, and relevant sales and stock data exists in the system.	1. Manager navigates to the Reports module and selects the report type (Sales, Stock, or Revenue). 2. Manager selects the required start and end date range. 3. System collects, calculates, and aggregates matching records. 4. System displays the report and allows export in CSV or PDF format.	Invalid or Reversed Date Range: System rejects the filter and displays a clear validation message.	The requested report is generated successfully and is available for viewing or download.
UC-08: View Low Stock Alerts	Manager	Reorder thresholds are configured for the products.	1. System monitors product stock levels after each inventory transaction. 2. When a product's stock falls below its configured reorder threshold, system identifies the product as low in stock. 3. System displays a low-stock alert banner on the dashboard. 4. Manager views the alert and can initiate a Purchase Order directly from the alert.	Pending PO Already Exists: System prevents duplicate auto-generated pending PO for the same product and informs the Manager.	The Manager is notified about products with low stock and can take immediate action to replenish them.
UC-09: Search and Filter Products	Staff	Staff is logged in, and products exist in the system.	1. Staff enters a search term or applies category and stock-status filters. 2. System queries the product database using the provided criteria. 3. System retrieves and displays matching products to the Staff member.	No Products Found: System displays the message 'No products found matching your search or filters.' with a Clear filters link.	Relevant products matching the search criteria are displayed to the Staff member.
UC-10: Login	All Users (Admin, Manager, Staff)	The user has a valid active account in the system.	1. User enters username and password. 2. System validates credentials against the database bcrypt hash. 3. System verifies the user's active status and assigned role. 4. System establishes a valid user session and redirects to the dashboard.	Invalid Credentials or Deactivated Account: System detects invalid credentials, displays 'Invalid username or password, or account deactivated.', and denies access.	The user is successfully authenticated and granted access according to their assigned role and permissions.

8. Requirements Traceability Matrix (RTM)

Table 8.1 maps all Functional (FR-001–FR-015), Non-Functional (NFR-001–NFR-012), and Security (SEC-001–SEC-016) requirements to their SRS section, architecture/code module, formal test cases, and verification status (P = Passed, PT = Partial, NT = Not Tested).

Req ID	Requirement Short	Section Ref / Design Spec	Code Module	Test Case(s)	Status	Comments
FR-001	Product CRUD & unique SKU	4.1 / DS-Inventory	app/presentation/routes/main.py, app/data/models.py	IT_03, IT_04, test_products.py	P	Verified via automated & manual tests.
FR-002	Auto stock update on sale/return/restock	4.1 / DS-Sales, DS-PO	app/business/sales.py, app/business/purchase_order.py	UT_01, IT_07, IT_08, IT_11	P	Verified across sale, PO receipt, and return.
FR-003	Low-stock alert below threshold	4.1 / DS-Inventory	app/data/models.py (Product)	ST_01, test_products.py	P	Dashboard banner & row badge verified.
FR-004	Create sale & generate invoice	4.2 / DS-Sales	app/business/sales.py, app/templates/invoice.html	UT_01, UT_02, IT_07	P	5% tax, subtotal, grand total, INV-<id> verified.
FR-005	User auth & RBAC (Admin/Manager/Staff)	4.3 / DS-Auth	app/presentation/routes/auth.py, app/presentation/authz.py	IT_01, IT_02, IT_09, ST_03	P	HTTP 302/403 enforced on protected routes.
FR-006	Add/edit supplier & link products	4.4 / DS-Supplier	app/presentation/routes/supplier.py	IT_09, test_suppliers.py	P	Admin-only supplier CRUD & product linking verified.
FR-007	Create PO & mark as received	4.4 / DS-PO	app/business/purchase_order.py	IT_08, test_purchase_orders.py	P	Pending → Received transitions & stock update verified.
FR-008	Search & filter products	4.1 / DS-Inventory	app/presentation/routes/main.py	test_products.py	PT	Functional filtering verified; <1s SLA not benchmarked.
FR-009	Sales, stock & revenue reports	4.5 / DS-Reports	app/presentation/routes/main.py	ST_02, ST_04, test_*_report.py	P	Date-range filtering & totals verified.
FR-010	Audit logging with user ID & timestamp	4.3 / DS-Audit	app/data/models.py (AuditLog)	IT_06, test_products.py	P	stock_edit, price_change, product_delete logged.
FR-011	Process returns/refunds & update stock	4.2 / DS-Returns	app/business/sales.py (process_return)	IT_11, test_sales.py	P	Stock incremented & ProductReturn recorded.
FR-012	Password reset for locked-out users	4.3 / DS-Auth	app/presentation/routes/auth.py, password_reset_email.py	test_auth.py, test_password_reset_email.py	PT	Signed token reset flow verified with mocked SMTP.
FR-013	Partial PO delivery & low-stock PO creation	4.4 / DS-PO	app/business/purchase_order.py	test_purchase_orders.py	P	Partially Received status & low-stock PO verified.
FR-014	Export sales & stock reports (CSV/PDF)	4.5 / DS-Export	app/business/report_exporter.py	test_report_exports.py	P	CSV and PDF byte output verified.
FR-015	Admin user create/edit/deactivate/reactivate	4.3 / DS-Auth	app/presentation/routes/auth.py	test_auth.py, test_auth_routes_coverage.py	P	Account lifecycle & self-deactivation block verified.
NFR-001	Sale processing ≤ 2 seconds	5 / DS-Perf	app/business/sales.py	UT_01, IT_07	PT	Fast local execution; no formal load benchmark.
NFR-002	99% uptime during store hours	5 / DS-Ops	Deployment / WSGI runtime	—	NT	Requires multi-week production telemetry.

Req ID	Requirement Short	Section Ref / Design Spec	Code Module	Test Case(s)	Status	Comments
NFR-003	bcrypt hashing & RBAC on all endpoints	5 / DS-Security	app/business/security.py, app/presentation/authz.py	IT_09, ST_03, test_security.py	P	Salted bcrypt ($2) and server RBAC verified.
NFR-004	New staff completes sale ≤ 15 min training	5 / DS-UX	app/templates/index.html	—	NT	Requires human usability trial.
NFR-005	Atomic transactions (no partial updates)	5 / DS-Data	app/business/sales.py, purchase_order.py	IT_10, test_sales.py	P	Rollback on mid-sale failure verified.
NFR-006	10,000 products & 50 concurrent users	5 / DS-Scale	Database & WSGI server	—	NT	Not load-tested with 50 concurrent users.
NFR-007	Modern browser compatibility	5 / DS-UI	app/templates/*, app/static/css/*	Manual UI verification	PT	Standard HTML5/CSS3; no automated multi-browser suite.
NFR-008	Modular layered architecture	5 / DS-Arch	app/presentation, app/business, app/data	test_app.py, full test suite	P	Three-tier package separation verified.
NFR-009	Automated daily DB backups (24h RPO)	5 / DS-Backup	scripts/backup_database.py	IT_12, test_backup_database.py	PT	Backup creation & integrity verified; cron/restore not tested.
NFR-010	Retain audit logs ≥ 6 months	5 / DS-Audit	app/data/models.py (AuditLog)	IT_06, test_products.py	P	6-month retention queryability verified.
NFR-011	HTTPS/TLS encryption in transit	5 / DS-Net	Reverse proxy / TLS termination	—	NT	Local tests use HTTP test client.
NFR-012	Clear, consistent error messages	5 / DS-UX	All route & service modules	UT_02, IT_02, IT_04, IT_05	P	Validation & flash alerts verified.
SEC-001..016	Supplementary Security Controls	5.1.2 / DS-Sec	app/business/security.py, auth.py, models.py	test_security.py, test_auth.py, test_products.py	PT	SEC-001, 003, 006, 009, 012, 013, 015 Passed; SEC-002, 005, 010, 011, 014 Partial; SEC-004, 007, 008, 016 Not Tested.